Before input validation

Prathyush Gowda HP / PES1UG24AM201 / Class 5-D

Step 2 / Run the starter application

The time interval produces 135 minutes. Entering bus stops the program at the rate lookup.

```
PS Lab 7> python parking_fee.py
==== Running minutes_parked ====
Enter entry and exit time (e.g. 09:30-11:45): 09:30-11:45
Minutes Parked: 135

==== Running hourly_rate ====
Enter vehicle type (bike/car/truck): bus
Traceback (most recent call last):
  File "C:\Users\prath\Documents\Codex\2026-11-12\https-github-com-setapesu26-l6-l7\outputs\
L6_L7_19_parking_fee\Lab 7\parking_fee.py", line 33, in <module>
    main()
  File "C:\Users\prath\Documents\Codex\2026-11-12\https-github-com-setapesu26-l6-l7\outputs\
L6_L7_19_parking_fee\Lab 7\parking_fee.py", line 24, in main
    result = hourly_rate(user_input)
             ^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\prath\Documents\Codex\2026-11-12\https-github-com-setapesu26-l6-l7\outputs\
L6_L7_19_parking_fee\Lab 7\parking_fee.py", line 10, in hourly_rate
    return RATES[vehicle.strip().lower()]
           ~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^
KeyError: 'bus'
PS Lab 7>
```

Step 4 / Run the completed Hypothesis tests

Six tests fail on the starter code. The existing rates for bike, car and truck pass.

```
PS Lab 7> python -m pytest -q --tb=no test_parking_fee.py
FFFF.FF                                                                  [100%]
================================= short test summary info =================================
FAILED test_parking_fee.py::test_minutes_parked_no_crash - ExceptionGroup: Hypothesis fou...
FAILED test_parking_fee.py::test_same_day_duration - AssertionError: assert -1439 == None
FAILED test_parking_fee.py::test_out_of_range_clock_fields - AssertionError: assert 1440 ...
FAILED test_parking_fee.py::test_hourly_rate_no_crash - ExceptionGroup: Hypothesis found ...
FAILED test_parking_fee.py::test_format_plate_no_crash - ExceptionGroup: Hypothesis found...
FAILED test_parking_fee.py::test_plate_formatting - AssertionError: assert 'AA  0 0  AA 0...
6 failed, 1 passed in 6.23s
PS Lab 7>
```

One failing interval, 23:59-00:00, returns -1439 minutes. An empty plate raises IndexError.

Both inputs are covered by the updated suite.

After input validation

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Step 5 / Repeat the application run

The same bus input now gives a message, and the program continues to the plate prompt.

```
PS Lab 7> python parking_fee.py
==== Running minutes_parked ====
Enter entry and exit time (e.g. 09:30-11:45): 09:30-11:45
Minutes Parked: 135

==== Running hourly_rate ====
Enter vehicle type (bike/car/truck): bus
Unknown vehicle. Choose bike, car or truck.

==== Running format_plate ====
Enter a number plate (e.g. ka01ab1234): ka01ab1234
Formatted Plate: KA 01 AB 1234
PS Lab 7>
```

Step 6 / Repeat the same tests

All seven tests pass against the fixed application.

```
PS Lab 7> python -m pytest -q --tb=no test_parking_fee.py
.......                                                                  [100%]
7 passed in 1.38s
PS Lab 7>
```

Validation rules

Time interval
Use 24-hour, same-day times. Equal times give zero minutes.

Vehicle rate
Trim whitespace and ignore case. Bike, car and truck cost 10, 30 and 60 per hour.

Number plate
Accept compact or spaced plates and return the uppercase form KA 01 AB 1234.

Each function returns None for invalid input. The command-line program explains the error.

Run environment

Windows PowerShell / Python 3.12.14 / pytest 9.2.0 / Hypothesis 6.170.15 / seed 201